

ZAINAB KHAWAR

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EDUCATION

Bachelor of Science in Information Technology (BSIT)

2022 – 2026

Bahria University, Islamabad

PROFESSIONAL SUMMARY

IT graduate with hands-on experience across AI/ML, backend development, and cybersecurity through three internships and a final-year cybersecurity platform (DRPE) built with FastAPI, JWT/RBAC, and automated vulnerability scanning. Comfortable building and shipping REST APIs, integrating ML models into production-style services, and working across the stack from Flutter mobile apps to SQL Server backends. Now working in software quality testing with DevOps exposure, and looking to keep building toward backend, QA automation, and AI-adjacent engineering roles.

SKILLS

Languages: Python, JavaScript, C++, SQL

Concepts: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming (OOP)

Web & Frameworks: React, HTML5, CSS3

Tools & Platforms: Git, Docker, Docker Compose, VS Code, Linux, Jira

Professional: Problem Solving, Analytical Thinking, Communication, Teamwork, Time Management

PROFESSIONAL EXPERIENCE

Python Intern — Evosoft

Oct 2025 – Dec 2025 | Remote

- Developed Python applications and backend services to support project requirements.
- Worked with REST APIs, databases, and data processing tasks.
- Assisted in debugging, testing, and optimizing application performance.
- Collaborated with team members to deliver project milestones on time, following version control and documentation best practices.

AI/ML Intern — DataCompass — Bahria Incubation Center

Jun 2025 – Sep 2025 | Islamabad

- Developed and integrated AI-based healthcare solutions using Python, TensorFlow, and Keras.
- Built and optimized a chest X-ray disease detection system using transfer learning (VGG16), evaluated on precision, recall, and F1-score.
- Integrated machine learning models with Flask-based REST APIs for inference and backend communication.
- Evaluated model performance using precision, recall, F1-score, and confusion matrices while addressing class imbalance.
- Collaborated with the development team to test, debug, and prepare the solution for deployment.

Web Development Intern — Insaafdaar

Jun 2024 – Sep 2024 | Remote

- Developed and enhanced responsive web interfaces using HTML, CSS, JavaScript, and React.
- Implemented new features and improved user experience across web applications.
- Collaborated with developers using Git and Agile development practices.
- Integrated frontend components with backend APIs and participated in testing and debugging.
- Contributed to code reviews and maintained clean, reusable, and efficient code.

PROJECTS

Dynamic Risk Posture Evaluation Using Automated Threat Intelligence (Final Year Project)

Technologies: FastAPI, PostgreSQL, OpenVAS, Nmap, JWT, Docker

- Developed a cybersecurity platform that automates vulnerability assessment and risk evaluation across networked endpoints.
- Built secure REST APIs with FastAPI, implementing JWT authentication and Role-Based Access Control (RBAC).
- Integrated OpenVAS and Nmap for automated vulnerability scanning.

- Developed a centralized dashboard to help organizations monitor and prioritize security risks.

MediScan AI — AI-Based X-ray Disease Detection System

Technologies: Python, TensorFlow, Keras, Flask, VGG16, Grad-CAM

- Developed an AI-powered healthcare application for chest X-ray disease detection using deep learning.
- Built and integrated a Flask REST API for model inference.
- Optimized model performance and evaluated results using precision, recall, and F1-score; addressed class imbalance to improve prediction reliability.

Task Management System

Technologies: React, ASP.NET Core Web API, SQL Server

- Designed and developed a full-stack task management application with secure user authentication and CRUD functionality.
- Integrated REST APIs with a SQL Server database and built a responsive UI for efficient task tracking.

SavorySync — Recipe Management Application

Technologies: Flutter, Firebase, Dart

- Built a cross-platform mobile application enabling users to browse, save, and organize recipes.
- Integrated Firebase for authentication and real-time data storage; implemented meal planning and shopping list features.

Car Racing Game

Technologies: C++, SFML/Graphics Library

- Developed a 2D car racing game featuring player controls, obstacle detection, score tracking, and collision handling.
- Applied object-oriented programming principles and game logic to build an interactive desktop application.

CERTIFICATIONS

- Foundations of Cybersecurity — Google (Coursera)
- Play It Safe: Manage Security Risks — Google (Coursera)
- Google AI Essentials — Google (Coursera)
- Data Science Foundations — IBM
- Advanced Python Programming — NETSOL Technologies
- Advanced Commands in Linux — Coursera
- Certified LLM Security Expert (CLLMSE) — Red Team Leaders

AWARDS & ACHIEVEMENTS

- 3rd Place — BUECS Open House 2026

LANGUAGES

English (Professional Working Proficiency), Urdu (Native)

INTERESTS

Reading, Travelling